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Seminar Report

User Experience Evaluation For Birzeit Linguistic Search Engine

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Abstract

More and more people are using the web for language research from spell-checking to finding academic papers, but using the existing search engines can be limiting and useless for linguistic research, which leads to wasted effort and distorted results. So the existing search engines cannot be depended on for reliable lexical results.

For this reason, a linguistic search engine at Birzeit University (Jarrar M, Amayreh H, 2019) was developed with high quality and rich resources to be used for translation service and also to introduce the concept of ontology (Jarrar M, 2019) to those who are not familiar with it.

8 years of hard work and determination to come up with this high quality linguistic search engine with more than 150 different digitized dictionaries (Jarrar M et al, 2019) that provide translation, synonyms, meaning and definitions.

This research aims to conduct a usability evaluation for this linguistic search engine, which is very important to understand the users' perception and the nature of the interaction with the website. Many criteria were considered to evaluate this search engines' user satisfaction measures (e.g. effectiveness, efficiency, learnability and design). A questionnaire was designed and used to collect the opinions of 620 users on the design and functionalities of the website. **Additionally a controlled experiment was conducted with 12 participants from two universities to measure effectiveness and efficiency.**

It was concluded that the result of measuring each criterion was satisfiable, with the effectiveness rating the most high average, also the rate of learnability was high in that the users found the search engine easy to use and understand.

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Chapter 1

Introduction

Usability evaluation is a very important aspect of any user interface to measure how easy the interface is to use, also evaluates the quality of a website and gives insights into potential problem areas. Many usability evaluation methods were developed to identify ways to improve the interaction with the system in order to increase its usability. Based on the identified problems, these methods can be categorized into: user based methods, which include the involvement of users to measure their performance and satisfaction of the tested interface. Also the evaluator based methods, which includes the involvement of experts to identify usability problems, like using heuristic evaluation to assess the user interface based to some heuristics. Where these two methods are the most used for usability evaluation. Finally, the tool based methods, which includes the involvement of software tools to assess the conformance to a set of usability guidelines (Hasan, L. 2012). Furthermore controlled experiment is very efficient to validate the results obtained from the user based evaluation (García, F.J. et al, 2007).

Furthermore, ISO standard defined usability as the extent to which users utilize a product effectively, efficiently along with satisfaction to achieve specific goals. However, the usability of a search engine can be defined somewhat differently. As to achieve the effectiveness, the search engine should return results relevant to users' entry. While efficiency can be measured from the number of searches that the user make to obtain the required results, and to achieve satisfaction, the search engine should have the ability to interpret the users entry in natural language to a defined search query (Taksa, I. et al, 2008).

On the other hand, to distinguish between usability and user experience, ISO standard defined user experience as the persons responses and perceptions that results from the interaction and/or anticipated use of a system, product or service, which is somehow similar to the concept of satisfaction in usability. Nonetheless there are some aspects of user experience missing in the definition like how it evolves from user expectation, then from full interaction with the system to form a whole experience that includes users' feedback and reflection on the system (Bevan, N., 2009).

Thus it was important to measure the user experience of the linguistic search engine to understand users' perceptions and the rate of conformance to their expectations about what a linguistic search engine is about.

What Birzeit's linguistic search engine and Arabic Ontology offer is more than just a simple translation of terms, but the search engine provides the feature of filtered results based on the users preferences, those filters are translation, synonyms, definitions and meaning. There is also the ontology feature to some concepts that introduce a unique knowledge to those who do not know about it. About 150 Arabic multilingual lexicons were collected, digitized, structured and stored in one database in order to provide a wide set of results (Jarrar M, 2019) (Amayreh et al. 2019).

So a user experience evaluation was carried out to get a hint of the users' feedback and expectations about it. The evaluation took many forms such as observation, interviews, questionnaire and a controlled experiment in a controlled setting, where it was done in laboratories and in natural settings.

1.1 Problem Statement

For an important and innovative search engine like this, there is no usability evaluation research conducted neither is there an evaluation for user experience and rate of satisfaction of the users who interacted and will interact with the search engine. Thus it is essential to understand users' preferences and perspectives about usability measures like effectiveness, efficiency, learnability and the interface design, in order to improve it to the best.

1.2 Research Objectives

The aim of this research is to conduct a usability evaluation of the linguistic search engine, In particular to measure user satisfaction and evaluate user experience through distributing a questionnaire and conduct a controlled experiment, in order to study and evaluate the different features of the search engine to achieve the following set of measures:

1. *Learnability*: by allowing the users to try the search engine and lead them through its beneficial features.
2. *Effectiveness*: Get users' feedback on the existing features and what they would like to improve by understanding their needs of accurate and sufficient results.
3. *Efficiency*: Get users' feedback on how the search engine helped in achieving the desired results fast and easy.
4. *Design*: by encouraging the users to give their opinion on the general design of the search engines user interface.

1.3 *Research Contribution*

Conducting a usability evaluation on the search engine, will add a great value towards its improvement. Thus performing a thorough analysis of the search engines features would help in solving any related problems.

A controlled experiment will be implemented to measure the usability criteria efficiency and effectiveness along with a user satisfaction questionnaire, and then the results will be analyzed in order to understand users' needs.

1.4 *Research Questions*

The research questions are derived from the criteria to be measured as follows:

RQ1. What are the design features that users find relatively comfortable?

RQ2. How easy and helpful do the users find the search engine along with its' different features?

RQ3. Can the users find what they are looking for fast and easy?

RQ4. Compared with familiar similar sites, do the users find the search engines' results accurate?

RQ5. Do the users find the results of the search engine sufficient?

1.5 *Overview of this research*

The research is organized in the following chapters;

Chapter 1: Introduction

This chapter presents an introduction of the usability evaluation and the motivation behind conducting this research, problem statement, research aims and objectives and contribution.

Chapter 2: Related Work and Background

In the following chapter, literature review on the subject from different areas will be presented. Then a review of different usability evaluation methods will be presented.

Chapter 3: Usability Evaluation

In this chapter the research methodology is discussed based on the data collection and data analysis with simple representative diagrams and a discussion of controlled experiment.

Chapter 4: Results and Discussion

In this chapter, the results and discussion of the questionnaire results will be presented along with the recommendations and the future work.

Chapter 2

Related Work and Background

2.1 *Usability Evaluation Experiments in Literature*

The usability of nine university websites in Jordan was evaluated by (Hassan L, 2012). Five categories were considered for the evaluation which are: content, navigation, design, architecture / organisation, communication and ease of use. A total of 237 students were selected to provide ratings for the developed usability categories. The selection of the nine universities was based on an international university ranking website, where the ranking of the selected ones was the highest. A pilot study was conducted beforehand to discuss the method of rating with the students and to explain any unclear points to them, then the results were considered for the actual survey. A pre-test survey was given to the students asking about their background and experience. Then each group of students were given ten minutes to explore three websites then rate each university on a 7-point Likert scale and answer two open ended questions regarding what they liked and disliked about the design features. The experiment took half an hour and was carried out in two weeks.

The results showed that in general the students were satisfied with the websites especially in the categories: ease of use and content, however some common weaknesses were concluded such as: The design of the sites, was mostly inconsistent, in terms of: fonts and colours, the general pages design, the difference in the Arabic and English language interface and the lack of support of Arabic language.

Another study was conducted also by (Hassan L, 2014) to get an insight on the point of view of students, in order to evaluate the design criteria of educational websites in Jordan. The students were asked to fill a pre-test questionnaire to study how some factors would affect the developed criteria of usability, factors such as gender and majors/specialisations were investigated. The results were analysed in several ways. Descriptive analysis was used for the pre-test questionnaire to find the weight for the developed criteria from the viewpoint of students; also the average weight was calculated. Furthermore ANOVA was used to calculate the variance for each category. The seven Likert scale was used to rate the different categories. The results showed dissatisfaction about the interface design of the selected websites, and satisfaction about how the websites were easy to use in terms of content and navigation.

(Mustafa, S.H. and Al-Zoua'bi, L.F., 2008) performed a usability study to figure out the strengths and weaknesses in the design of Jordanian Universities through studying many aspects of usability. Two evaluation methods were conducted, the questionnaire method and the method of using automated tools (html toolbox and web page analyse). The reason behind using two methods is that by using the automated tools, many internal features could be measured without the user involvement, while using the questionnaire method is for measuring 23 usability criteria to measure five categories, readability and organization of content; links and navigation; educational information; user interface design; performance and effectiveness.

Each category was given an index, and through some calculations, the usability level was determined. A total of 252 student responded to the questionnaire.

The results showed that the overall usability was satisfiable, with the first two categories, content and organisation and readability, achieved the highest usability level, thus they proved that using heuristic evaluation is the best method for usability evaluation. However, the results did not provide details about usability problems that users identified.

(Gul, H. and Saqib, M., 2015) also performed two kinds of heuristic evaluation, survey and expert review. Also a heuristic guideline document was developed to divide the heuristics into five major categories: Content, Design, Navigation, supporting the user tasks and Ease of Use. Furthermore a questionnaire was formed and distributed to the undergraduate students of one of the universities in Saudi Arabia. The results showed that the design and ease of use categories violate a number of heuristic rules. Many problems were also identified such as inconsistency in the interface language also figures are not aligned properly in the interface and the design was found complex for an educational website. Nonetheless, the authors did not offer any suggestions that can be employed to help improve the usability.

(Taksa, I. et al, 2008) implemented usability evaluation on a Meta search engine to measure the usability factors using many tasks of information retrieval. The experiment took the form of between-subjects design experiment. Three groups of 30 students each were given a paragraph as an example to form search queries for each group's task and retrieve relevant results. The experiment was conducted in three parts, each was adding to the previous additional tasks. The first, Conventional benchmark, in which students were asked to create search queries for the given paragraph and to rank all selected results using a scale from 1 to 9 on how relevant they were. The second, meta search engine for long query, where the students repeat

what they have done in the first part but with using LQMSE which is short for (Long Query Meta Search Engine) and finally the third part; meta search engine for motivated long query, where they were asked to repeat the second part with new conditions as to select the most relevant results and using the new search engine without previous knowledge and experience in it.

The results showed that regarding the LQMSE it would be helpful to offer assistance through on-line or context-sensitive help for the users especially when they are looking for lengthy information retrieval process. Also give the users control over the search and retrieval of results. Furthermore the usability results demonstrated that the learnability measure gives an overview about how users like to try new and experimental functionalities, and the measure of completeness showed a need for well-constructed system to help new users complete their searches.

(García, F.J. et al, 2007) performed a control experiment on a website called e-fashion, the goal of study was to validate different quality patterns, and thus they developed six versions of the website. The chosen quality criteria were “Guide” which includes: grouping, prompting, legibility and immediate feedback. Furthermore, each version of the website was assigned to a group of participants who were high school students, and each group was assigned four tasks. Afterwards they were asked to fill in a satisfaction test. Usability factors were measured: effectiveness, efficiency, satisfaction and learnability, by measuring the task completion, task time and error frequency. The results of each group were compared for each task and the mean value was calculated for the task time factor. The results showed that using patterns to describe the quality can produce better web applications; also the use of

satisfaction test can give a good overview of the users' perception and understanding of the website.

2.2 Usability and User Experience

A critical review of 206 research papers can be found in (Fernandez, A. et al, 2011) about usability evaluation methods through a mapping study to evaluate how such methods affect the website development process. Papers which were included in the study concerned using the definition of user evaluation methods applied for the web. In order to evaluate the quality of the selected papers, a questionnaire with a three point Likert scale was formed. The results showed that 39% of the studies used usability evaluation methods specifically user testing methods for web applications, moreover, 90% of the studies use the evaluations methods in the implementation phase of web development process, which is considered costly to perform change in this phase.

Another study by (Finstad, K., 2010) reviewed the usability metric for user experience which is a four point Likert scale to provide a subjective assessment of usability. The study conducted at Intel Company that relies on the system usability scale (SUS) which is 1 five point Likert scale with scoring system of weights 1-100 for its system usability evaluation. A total of 42 participants were asked to evaluate two systems, one with good usability and the other of bad usability. Between subject design was used to evaluate the two systems and then responding to the two usability metrics. It's been concluded that the usability metric for user experience is a valid, reliable and a sufficient usability measure.

(Vermeeren et al. 2010) presented a comparison between user experience and usability, has been found by many researches that usability addresses objective measures like task completion time and the rate of errors, whereas UX focus on the subjective measures like user feelings about the system, in other words it focuses on lived experience rather than task performance. It can be seen that the satisfaction factor in the usability is part of UX, but it addresses much more subjective qualities like user's motivation and expectations. Furthermore, 96 UX evaluation methods were collected and categorized based to some attributes

(Bevan, N., 2009) also established a solid comparison between the concept of user experience and usability. First by the objective differences, as the usability objective is to improve users performance while the UX objective is to improve user satisfaction to achieve realistic goals. Many aspects were considered to distinguish the differences, as user experience can be summarized to include an improvement to the satisfaction concept of usability, combining the perceptions and responses of users, whether measured subjectively or objectively.

Review of Relevant Background

2.3 *Linguistic Portals*

2.3.1 **Panlex**

A project by the Long Now Foundation (Kamholz et al., 2014) that aims to adapt and preserve the global linguistic diversity. It was started in 2005 as a research project at the University of Washington Turing Center. A lexical database was developed to cover literary and colloquial varieties of natural languages, as well as extinct, reconstructed, artificial, and controlled languages. Most of the data are based on bilingual, multilingual, and monolingual publications, such as dictionaries, word lists, Wiktionary, WordNet, and thesauri. It is used in many applications where It is particularly useful in author-machine collaborative translation, like the prototype Pan Images application based on Pan Dictionary, that converts the users search into many different languages to use it for image-search services, in order to discover relevant images labeled in languages they do not know, sometimes selecting particular target languages. (Baldwin et al., 2010)

2.3.2 **Arabic Wordnet**

The building of Arabic WordNet (AWN) (Elkateb et al. 2006) followed the same general methods developed for Princeton WordNet (PWN) (Fellbaum 1998) and EuroWordNet (Vossen 1998). Before releasing AWN, there was a great obstacle finding linguistic resources efficient enough for large scale Arabic NLP systems. After its release, AWN quickly became popular in the Arabic NLP community as a free available lexical and semantic resource (Abouenour et al. 2013). This unique

resource followed the WN structure, but taking into account some specificities of Arabic such as entry vocalization, Broken (irregular) Plurals (BP) and roots. Moreover, enabling translation on the lexical level to English and many other languages. (Elkateb et al. 2006)

2.3.3 Birzeit Linguistic Search Engine

Birzeit University launched a linguistic search engine with 150 multilingual lexicons and a linguistic ontology for Arabic, September 25, 2018. The search engine contains the world's largest Arabic lexical database, which is a non-profit, long-term research project that aims to provide people with an innovative tool to find translations, synonyms, and definitions in all domains and disciplines.

The Arabic ontology, which is a classification and a tree of the meanings of Arabic terms, is based on state-of-the-art science, rather than the native speakers' knowledge as in dictionaries. The main aim is to build a comprehensive linguistic data graph, which will include syntactic, morphological, and semantic features, as well as linking these to other languages. It is also represented in logic and as such can be understood by machines and used in modern applications, such as semantic information retrieval, machine translation and semantic disambiguation, big data and information integration, the Semantic Web, and others.

2.4 *Usability Definition and Basic Concepts*

Usability is considered the most traditional discipline in the human-computer interaction field, which studies the development of user interface, also as defined by ISO as a “measure of effectiveness, efficiency and satisfaction with which the product can be used by specified users to achieve certain tasks in a specific context” (Standard ISO 9241-11, 1998). According to Jakob Nielsen (2012), a leading professional in HCI, usability is defined by 5 quality components:

- **Learnability:** How easy is it for users to perform basic tasks the first time they use the system?
- **Efficiency:** How long it would take them to perform tasks?
- **Memorability:** After a period of time of not using the system, how easily can they remember the tasks the performed before?
- **Errors:** How many errors do users make, know its severity level, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

Similarly ISO/IEC 9126-4 states that usability metrics should include:

- **Effectiveness:** The accuracy and completeness with which users achieve specified goals
- **Efficiency:** The resources used in relation to the accuracy and completeness with which users achieve goals, typical resources include time, human effort, costs and materials
- **Satisfaction:** The comfort and acceptability of use where the user experience that results from actual use meets the user’s needs and expectations.

Additionally, product effectiveness as defined by (Simpson and Weiner, 1989) is "the degree to which something is successful in producing a desired result". Effectiveness can be measured by many parameters, such as implementing many user tasks in the product, the rate of successful actions made by the user, building an intuitive user interface and perform as many features and commands as possible.

Product Efficiency can be described as “the resources that a user spends to accurately and completely achieve their goal”. Efficiency is measured by many parameters such as the minimum time to perform a task, minimize the time to train the users on using the system, not waiting too long to respond back from the system when entering the system and easily get the required information from the system.

Product Satisfaction is defined as "the number of customers, or percentage of total customers, whose reported experience with a firm, its products, or its services (ratings) exceeds specified satisfaction goals" (Farris, 2010, 40). Satisfaction is measured by many parameters such as: The system user interface should meet the expected usage, the output of the system as expected and the user is satisfied for the use of system.

2.5 *Usability Evaluation*

Evaluation of any website depends on both the users' experience when interacting with the system that should be satisfying, enjoyable and motivating and the usability of the system as of how easy it is to learn and use the system. Evaluation gives the opportunity to check whether the design is appropriate and acceptable for the targeted population (Preece, J, Rogers, Y, Sharp, H. 2015).

Different evaluation methods exist that depends on the goal of the evaluation and the environment where the evaluation take place as in laboratories, people's homes, outdoors, and work settings.

2.5.1 Types of Evaluation

2.5.1.1 Controlled settings

This type involves collecting data by using many methods such as experiments, observation, interviews, and questionnaires. It is conducted in many settings like in laboratories or in natural settings. It aims to investigate how typical users perform on basic tasks, decide if an interface is usable by the targeted population, compare the number and kinds of errors that the users make and how long it would take them to complete a task (Preece, J, Rogers, Y, Sharp, H. 2015).

2.5.1.2 Natural settings

This setting involves users, such as field studies and in the wild studies to understand how the product will perform in the real world and to understand the users behavior with the new technology (Jambon, F., Meillon, B. 2009). It is used for technologies being developed for the use outside office settings like, mobile and other technologies which are ready to use in the home, outdoors, and in public places. It aims to identify opportunities for new technology, eliciting requirements for new design and what is the best way to introduce a new technology and evaluate technology already in use.

2.5.1.3 Settings not involving users

In this type, there are no users so the researcher has to imagine or model interface to be used, this takes many forms like, inspections, walkthroughs, models, heuristics and analytics. It aims to identify usability problems, concerning users' behavior, the conditions in which the system will be used, and the kinds of actions performed by the users. One of the problems of using this setting is that the findings are not very accurate (Preece, J, Rogers, Y, Sharp, H. 2015).

2.6 *User-Based Usability Evaluation Methods (UEMs)*

This type of evaluations is the most common to be used to evaluate the usability of websites.

2.6.1 User Testing

The user testing method is “a systematic way of observing actual users trying a product and collecting information about the specific ways in which the product is easy or difficult for them” (Dumas and Redish, 1999). It is the most useful approach as it provides a general understanding of how real users use the interface and the kinds of problems the users make in their interaction.

User testing is conducted in many ways, such as observation (e.g., notes, audio, video or interaction log files) to measure the performance of the user; questionnaires and interviews to measure users' satisfaction (Rogers et al., 2011).

2.6.2 Think-Aloud Method

This is a type of user testing method, but with asking users to think aloud during their interaction with an interface (Lazar, 2006). In order to understand how users interpret with an interface. However, this method down side is that the test setting does not represent a natural setting as it needs a recording equipment; therefore this will not encourage users to act or talk naturally (van den Haak and de Jong, 2005).

2.6.3 Constructive Interaction

This is a type of think-aloud method, but instead of a single user, it needs two users interacting with an interface together to complete specific tasks (Holzinger, 2005). This offers more natural settings compared with the Think-Aloud tests, as people tend to verbalize what they think with others.

2.6.4 Questionnaires and Interviews

Different types of questionnaires (e.g., closed or open) and interviews (e.g., unstructured, semi-structured or structured) are a useful and simple way to measure users' satisfaction about a user interface, in terms of the features and presentation of websites (Rogers et al., 2011). These could be used before the user testing method or alone and this case they are considered as indirect usability methods as they would only reflect users' opinions about that interface (Holzinger, 2005)

2.6.5 Eye Tracking

In this type, a special eye tracking equipment is needed and built into the computer monitor with an eye tracking software to track the user's screen. It aims to record and observe the exact paths the users follow on a computer screen while using

the web (Nielsen and Pernice, 2010). This method is useful to observe normal users behavior and can discover exactly what users look at. However, this information cannot be accurate to measure if users are happy or confused when they look at certain things on a screen. Furthermore the special equipment required for this method is expensive.

Chapter 3

Usability Evaluation and controlled Experiment

In this chapter, the research methodology is described in terms of what methods were carried out, the survey design guidelines, targeted population, the implementation of controlled experiment and how the data was collected and analyzed in order to provide information about the research questions.

3.1 *Targeted Users*

1. Students searching for translation
2. Normal people (above 18 years) searching for meaning
3. Writers (above BA degree) searching for synonyms
4. Researchers (senior students and above) searching for meaning/synonym/ontology.

3.2 *Research Objectives Establishment*

In this step, the research objectives were established to identify the impact of the users when they interacted with the Linguistic search engine and then answering the questionnaire in order to measure specific usability criteria.

3.3 *Survey Design*

According to (E. Martin. 2006), the flow of the questionnaire including wording and ordering decision, formatting and response categories, construct is the key that represent another form of indirect research question.

The questionnaire was formed to collect data about the users experience with the linguistic search engine. Then the collected data was used to assess a set of specific criterion.

Our survey consists of 16 questions that were carefully put together to cover all the aspects regarding the usability evaluation. All the questions were designed to be multiple choice questions in order to get better responses. The following points were investigated: what is the aim of using the search engine, whether the general design of the search engine was comfortable and attractive to the user, if the search engine was easy to use, if the speed of the search engine was acceptable and many more points will be explained later on.

3.4 *Data Collection*

To collect data for this research, more than 1000 questionnaires were distributed to the targeted population for the period of 2 months. The total number of responses acquired was 622.

Two pilot questionnaires were formed and distributed to a small group of users to get a feedback on their aesthetic preferences, and most of them chose the questionnaire that is formed on the google forms, thus it was the one that was then distributed.

The questionnaire was either distributed during lectures after getting the permission from the lecturer or students were selected randomly outside the lecture and different users from work or researchers from different fields.

First, they were asked if they heard about the Linguistic search engine, in fact so many knew and heard about it already, and then they were asked to enter the search engine from the computer or mobiles. The nature of the search engine and different features were demonstrated and they were led to use and interact with the it and try as many words as they like, despite the fact that the time in the lecture was restricted, thus the students were given less time to try so many words. Some asked for more explanation about the filters and gave so many verbal or written notes. Afterwards, they were asked to fill the questionnaire.

The first part of the questionnaire focused on the different user types and age, the aim of using the search engine and how many words the user tried before answering the questionnaire.

The second part concentrated on measuring the different criteria of the usability evaluation which will be analyzed in the next section.

3.5 *Data Analysis*

The questions were analyzed with each answer given a percentage in order to know the importance of each option compared to other options. Additionally, the average of each question and its embranchments was calculated and rated according to the rating table provided by (Mustafa, S.H. and Al-Zoua'bi, L.F., 2008) as shown in table1; however these ratings are subjective and reflects users' choices and preferences.

Points (x)	Usability Level
$0 < x < 20\%$	Bad
$20\% < x < 40\%$	Poor
$40\% < x < 60\%$	Moderate
$60\% < x < 80\%$	Good
$80\% < x < 100\%$	Excellent

Table 1: Usability points and the corresponding usability levels provided by (Mustafa, S.H. and Al-Zoua'bi, L.F., 2008)

The first part of the questionnaire included four questions to understand the user's background, age, what are they looking for in the website and how many words they tested after interacting with the linguistic search engine. These questions were set as variable factors, to see how changing their results can effect on the results of each criterion.

The second part of the questionnaire included 12 questions which were scored on a 4-point Likert scale on strength of agreement according to (Finstad, K., 2010) based on the usability metric for user experience which provides a subjective assessment of usability. The questionnaire was set to measure usability criteria: design, learnability, efficiency and effectiveness.

3.5.1 Type of respondents

Most of who responded on the questionnaire was students as shown in figure 1, as we targeted university students from Birzeit University and Kadoori technical University to be our most targeted population, since it was intended to introduce the ontology search engine to them as it would be a great beneficial site to help them with

their assignments and homework's. Many employees from both universities responded as well, along with some researchers and writers from different faculties.

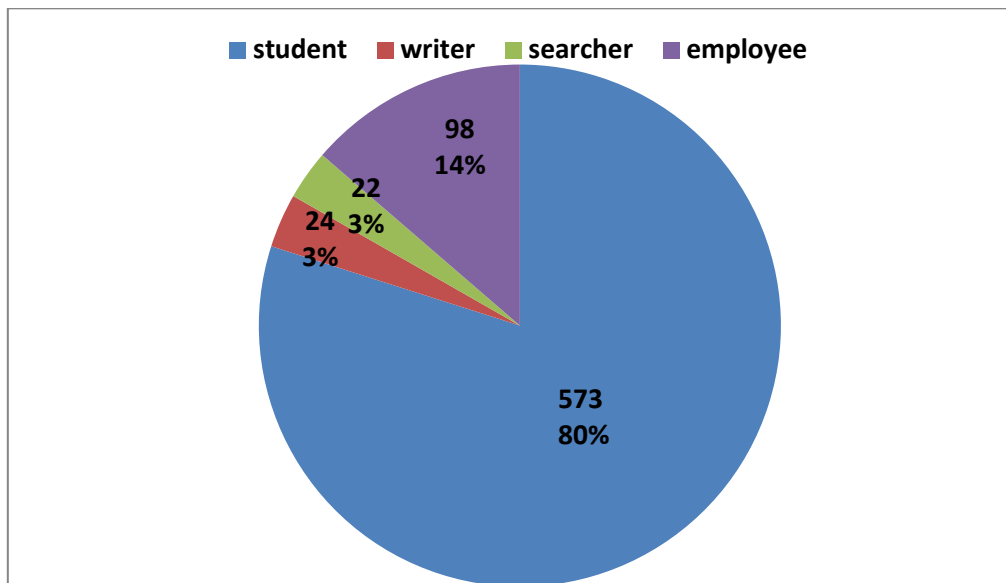


Figure 1. Types of respondents

3.5.2 Respondents age

As shown in figure 2, most of the respondents are of age between 18 and 25 as all of them are students and the rest of the respondents are master students, researchers or employees.

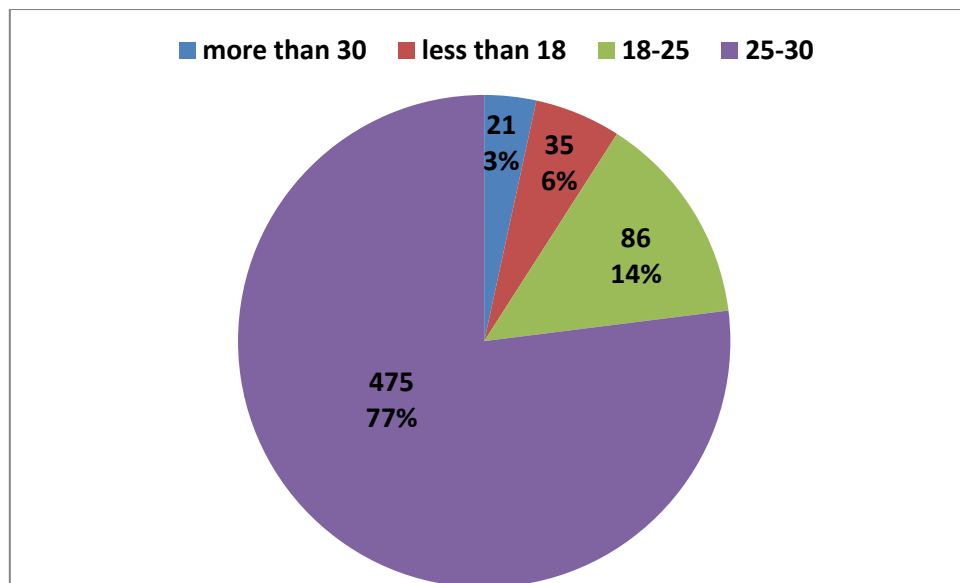


Figure 2. Respondents' age

3.5.3 The aim of using the Birzeit Linguistic search engine

Three options were given as to understand the main aim of the respondents for using the search engine based on the existing filters in the site, as shown in figure 3, most of the respondents (49%) picked “meanings and definitions” as their answer, followed by “looking for translations” with (32%), then ”looking for synonyms” with (19%)

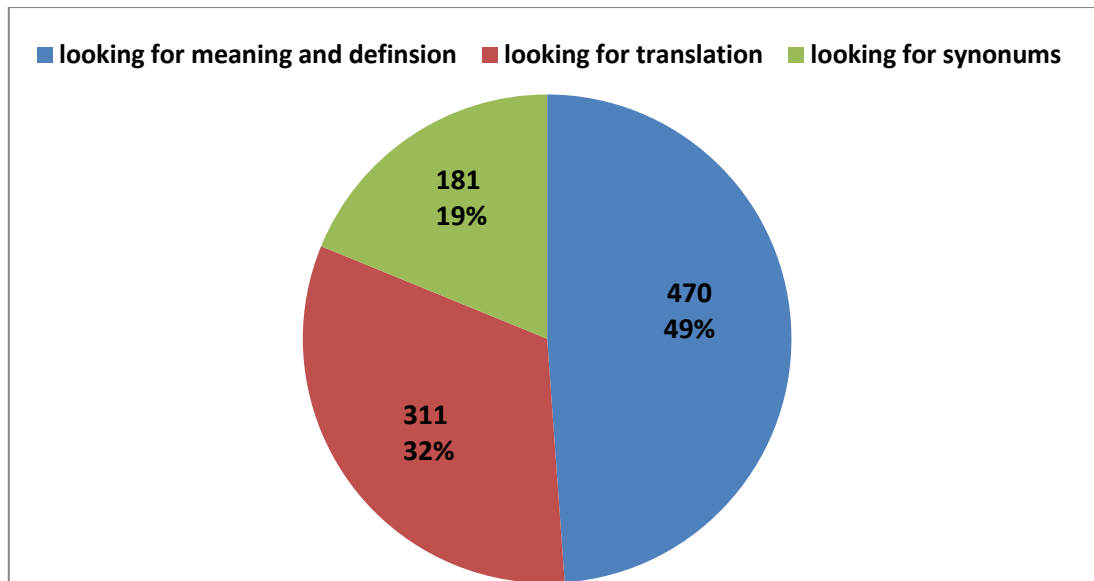


Figure 3. The aim of using the Birzeit Linguistic search engine

3.5.4 Number of searched words

Before answering the questionnaire, the respondents were given some time exploring and interacting with the Ontology search engine's functionalities through looking for some words, the reason behind the high percentage of the option "one word" and " five words" is that most of the experimented students were during lectures, so there was not enough time to try more than five words.

As shown in figure 4, (283 respondent) chose the option "one word" with (46%), then came the second option "more than five words" with (270 respondent) (44%) and lastly the option "more than 20 words" with (64 respondent) (10%).

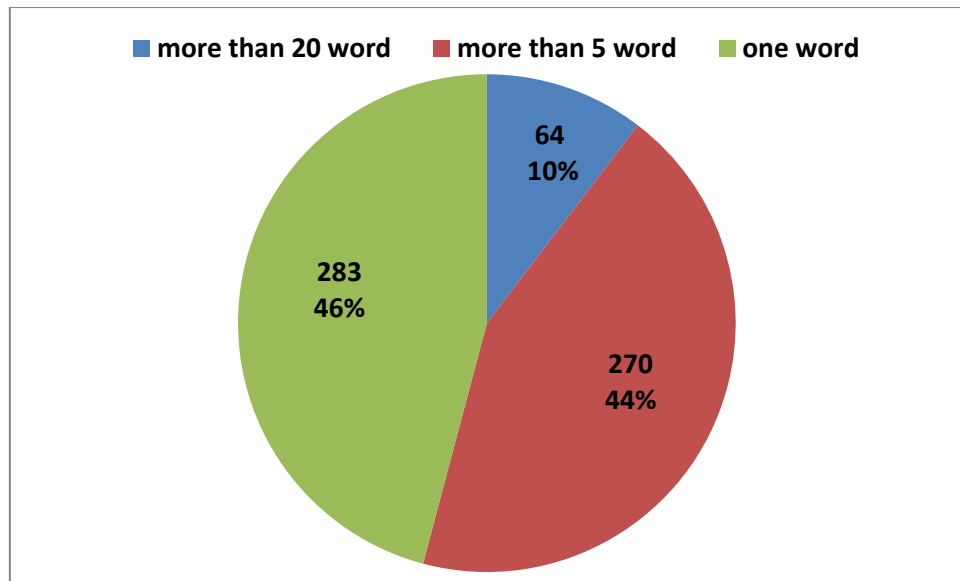


Figure 4. Number of searched words by each user

3.5.5 Fonts and colors of Birzeit Linguistic search engine

The first question in the evaluation process of the search engine usability is how attractive and comfortable the general colors and fonts, figure 5 shows the answers of all the respondents, where most of them answered “good” with (317 respondent), then the answers “excellent” and “acceptable” with nearly same values with (137 respondents) and (127 respondents) respectively. The average of this question is 2.92\4 (73%).

Figure 6 shows all the responds excluding the users who answered the option “one word”, the majority answered “good” with (175 respondent), then “excellent” (75 respondent), “acceptable” (58 respondent) and “weak” with (19 respondent). The average of this question is 2.94\4 (73.5%)

The average of both options can be considered satisfiable as people tend to have different perceptions and preferences, so satisfying more than 73% can be considered really good.

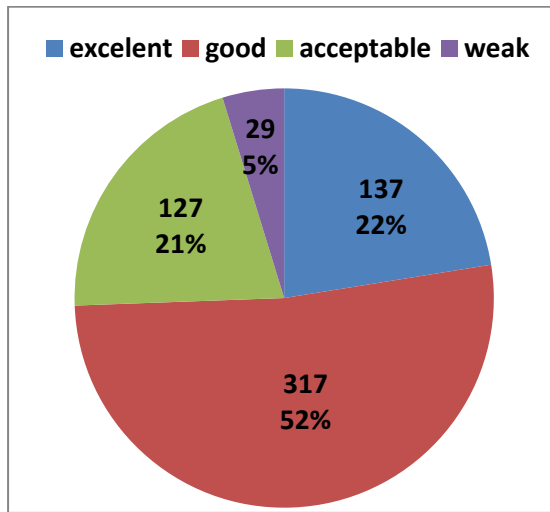


Figure 5. all users opinion about fonts and colors of the search engine

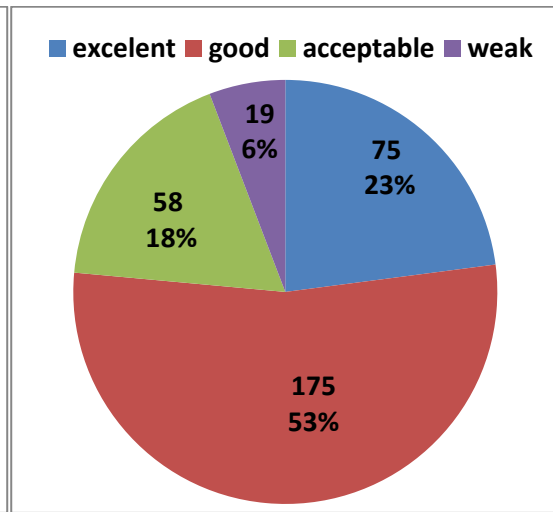


Figure 6. Users who tried more than one word opinions about fonts and colors of the search engine

3.5.6 The presentation of results

This question focused on how comfortable the presentation and the arrangement of the results is for the respondents, figure 7 shows the answers of all the respondents, where most of them answered “good” (304 respondent) then “excellent” with (145 respondent) and “acceptable” with (114 respondent) and last “weak” with (51 respondent). The average of this question is 2.92/4 (73%).

Figure 8 shows all the responses excluding the users who answered the option “one word”, the majority answered “good” with (165 respondent), then “excellent” (72 respondent), “acceptable” (66 respondent) and “weak” with (31 respondent). The average of this question is 2.83/4 (70.75%).

Respondents gave different feedback regarding this criterion, such as the ontology box on the right is confusing, the results should be presented in the middle of the page. Each tried to conform the presentation of results to their aesthetic preferences. Despite that the average of both options can be considered satisfiable.

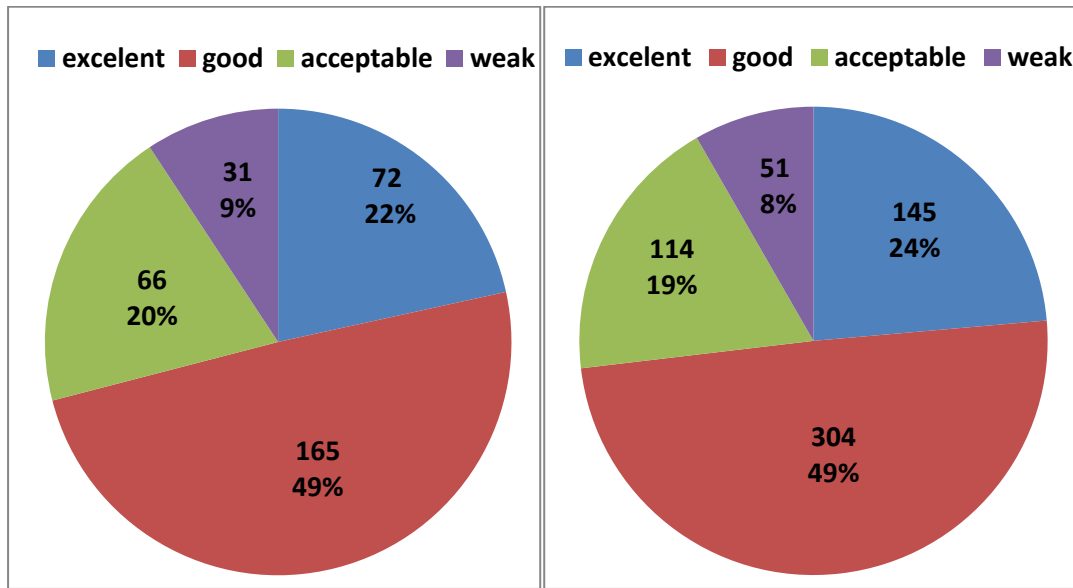


Figure 7. all users opinions about the Presentation of results.

Figure 8. Users who tried more than one word opinions about Presentation of results engine

3.5.7 The amount of results

In this question the respondents were asked: how comfortable the amount of the results in a single page, figure 9 shows the answers of all the respondents, where most of them answered “good” (288 respondent) then “excellent” (179 respondent) and “acceptable” (117 respondent) and last “weak” (30 respondent). The average of this question is $\frac{3}{4}$ (75%).

Figure 10 shows all the responses excluding the users who answered the option “one word”, the majority answered “good” with (151 respondent), then “excellent” (81 respondent), “acceptable” (78 respondent) and “weak” with (21 respondent). The average of this question is $\frac{2.88}{4}$ (72%)

Respondent were divided with their opinions, as some considered that ten results in a single page is good, while others considered it too much as they didn’t

want to scroll down for more than five results in a single page. Again this goes back to how each person prefers, but the total average is satisfiable.

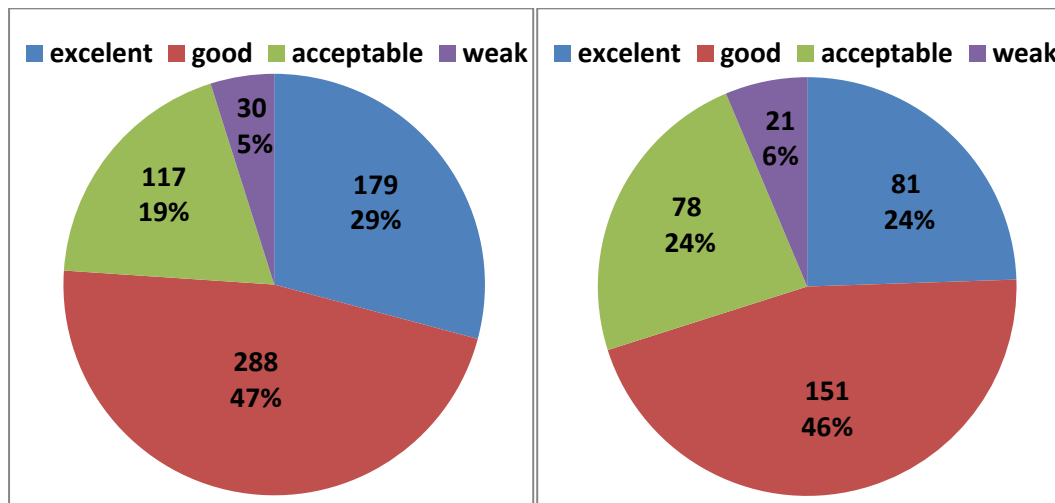


Figure 9. all users opinions about the amount of results.

Figure 10. Users who tried more than one word opinions about amount of results engine

3.5.8 The easiness of the linguistic search engine

This question focused on how easy it was for the respondents to use and learn the functionalities of the search engine, figure 11 shows the answers of all the respondents, the answers “excellent” and “good” were nearly the same with (288 respondent) and (252 respondent) respectively, then “acceptable” with (60 respondent). The average of this question is 3.3/4 (83%).

Figure 12 shows all the responds excluding the users who answered the option “one word”, the majority answered “excellent” with (155 respondent), then “good” (137 respondent), “acceptable” (29 respondent) and “weak” with (12 respondent). The average of this question is 3.3\4 (83%).

Figure 13 shows only responses of the students, to see how east it was for the students to use the linguistic search engine for the first time , the majority answered

“excellent” with (269 respondent), then “good” (233respondent) and “acceptable” (56respondent). The average of this question is 3.3\4 (83%).

Users were satisfied that the search engine is so easy to use and comprehend and that all the functionalities and features are obvious and memorable, especially for students, who got to know and interact with the search engine for the first time, their average came high and satisfiable.

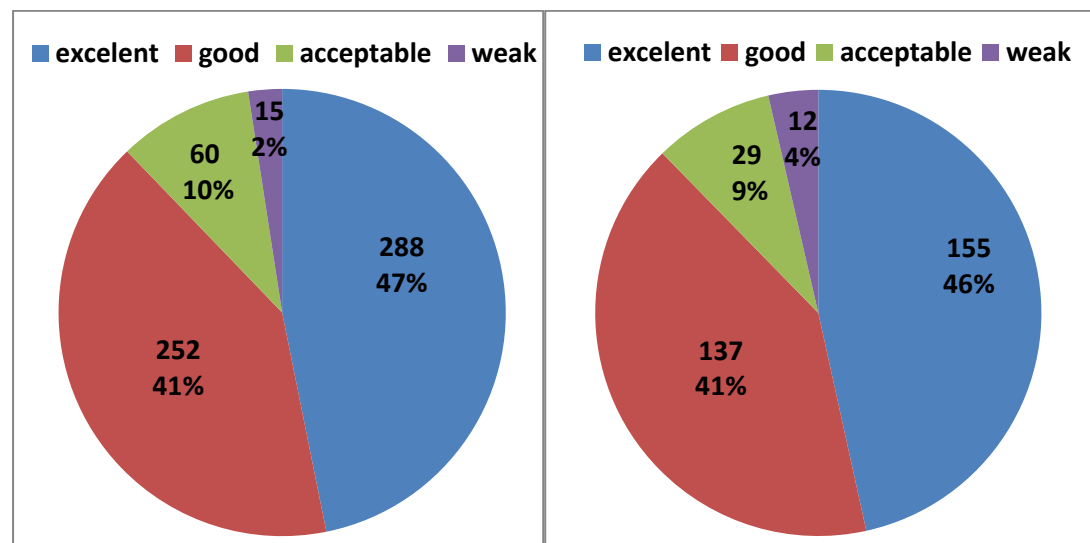


Figure 11. all users opinions about how easy is the Linguistic search engine

Figure 12. Users who tried more than one word opinions about how easy is the search engine

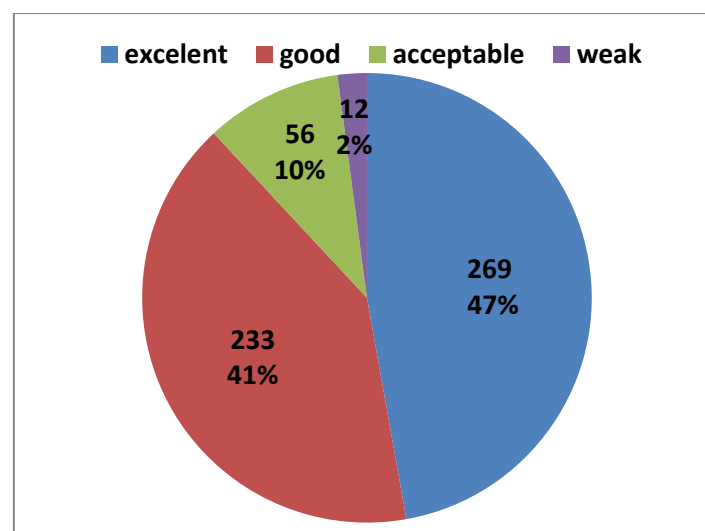


Figure 13. students opinions about how easy is the search engine

easy is the Linguistic search engine.

3.5.9 The functionality of the filters

In this question the respondents were asked if they found the filters easy to use and if it was easy to understand how they work, figure 14 shows the answers of all the respondents, where most of them answered “good” (276 respondent), then “excellent” (191 respondent) and “acceptable” (110 respondent) and last “weak” (37 respondent). The average of this question is $3\frac{3}{4}$ (75%)

Figure 15 shows all the responds excluding the users who answered the option “one word”, the majority answered “good” with (147 respondent), then “excellent” (102 respondent), “acceptable” (62 respondent) and “weak” with (21 respondent). The average of this question is $3\frac{3}{4}$ (75%).

Figure 16 shows only responses of the students, to see how east it was for the students to use the linguistic search engine for the first time , the majority answered “good” with (216 respondent), then “excellent” (164 respondent), “acceptable” (86 respondent) and “weak” (28 respondent). The average of this question is $3\frac{3}{4}$ (75%).

Figure 17 shows only the response of the users who chose the option “looking for translation” as their aim of using the linguistic search engine, the majority answered “good” with (255 respondent), then “excellent” (178 respondent), “acceptable” (103 respondent). The average of this question is $3.2\frac{1}{4}$ (80%).

When it came to using the filters, users had no problems in using the filters and they found them easy to use and understood the differences between them, and it shows in the high average of the question where users chose the option “looking for translations”. The total average was very good, especially the average of the students came high.

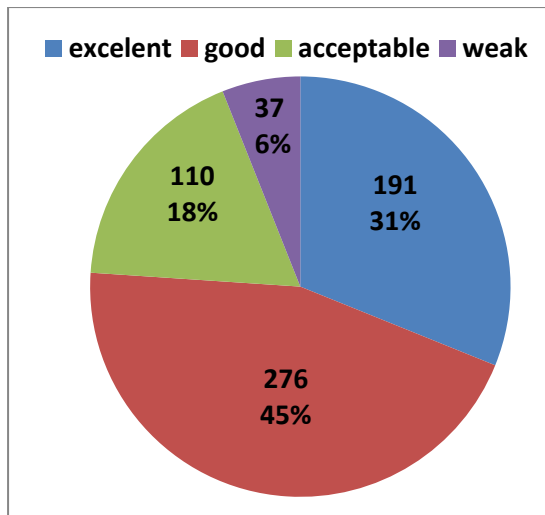


Figure 14. all users opinions about how easy is to use the filters.

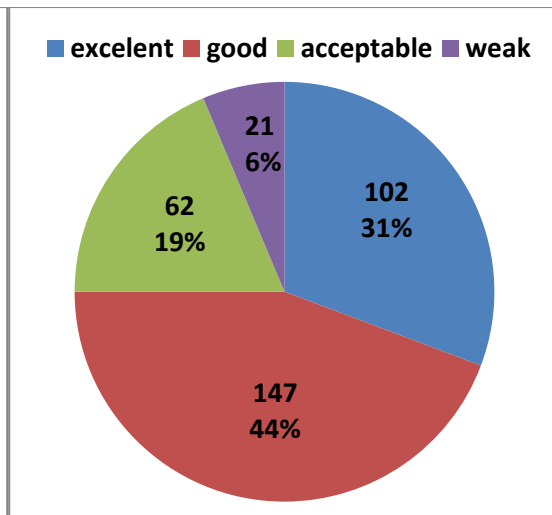


Figure 15. Users who tried more than one word opinions about how easy is to use the filters

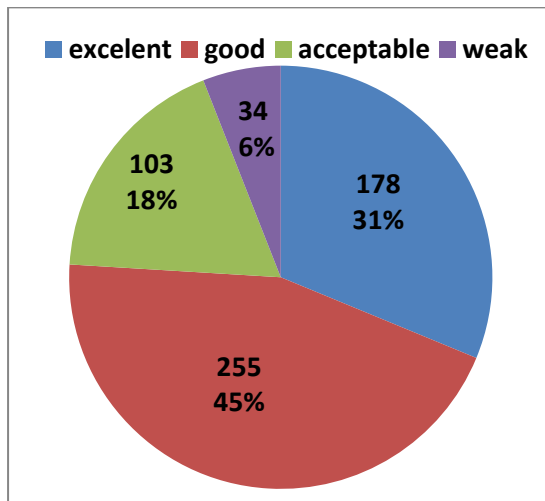


Figure 16. students opinions about how easy is to use the filters.

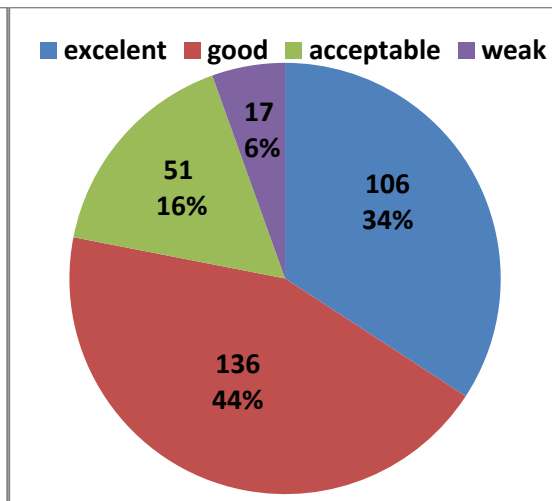


Figure 17. Users looking for translation opinions about how easy is to use the filters

3.5.10 The speed of the Arabic Ontology search engine

The respondents were asked to rate the speed of the search engine, figure 18 shows the answers of all the respondents, most of them answered “excellent” (322 respondent) and this indicates that the speed is very good and responsive, then they answered “good” (221 respondent) and last with nearly same values “acceptable” and

“weak” with (50 respondent) and (23 respondent) respectively. The average of this question is 3.3\4 (82.5)

Figure 19 shows all the responses excluding the users who answered the option “one word”, the majority answered “good” with (175 respondent), then “excellent” (122 respondent), “acceptable” (28 respondent) and “weak” with (9 respondent). The average of this question is 3.4\4 (85%).

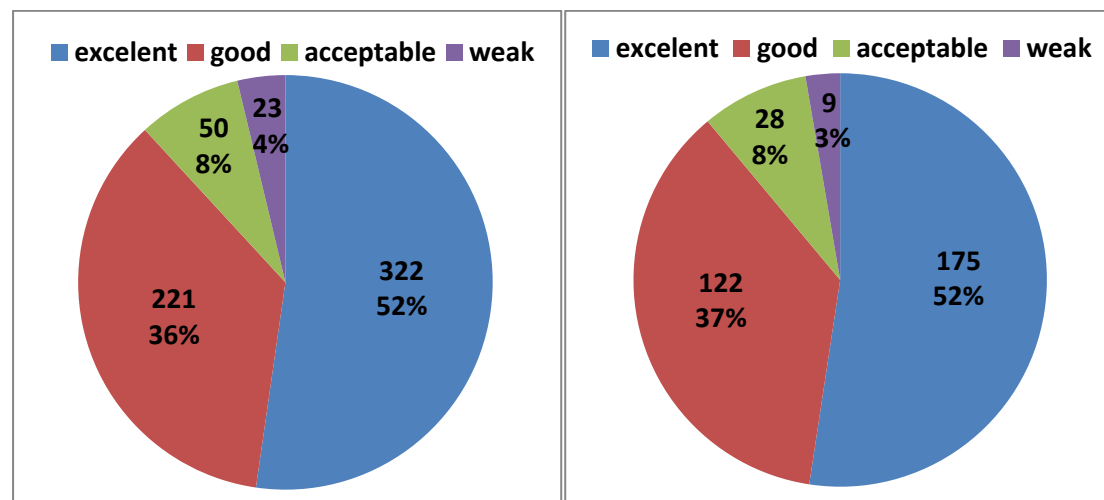


Figure 18. all users opinions about the Speed of the Linguistic search engine.

Figure 19. Users who tried more than one word opinions about the speed of the search engine

3.5.11 Helpfulness of the filters

The respondents were asked if the filters help in reaching the desired results quickly and efficiently, figure 20 shows the answers of all the respondents, most of the respondents answered “good” (309 respondent), while others responded “excellent” (171 respondent), then (97 respondent) and (37 respondent) for “acceptable” and “weak” respectively. The average of this question is 3\4 (75%).

Figure 21 shows all the responds excluding the users who answered the option “one word”, the majority answered “good” with (172 respondent), then “excellent”

(90 respondent), “acceptable” (52 respondent) and “weak” with (20 respondent). The average of this question is 3¼ (75%).

Figure 22 shows the responses of all the users excluding the students, to see their opinions regarding the functionality of the filters in the linguistic search engine, the majority answered “good” with (19 respondent), then “acceptable” (4 respondent) and “excellent” (3 respondent). The average of this question is 2.9¼ (72.5%).

Figure 23 shows only the response of the users who chose the option “looking for translation” as their aim of using the linguistic search engine, the majority answered “good” with (174respondent), then “excellent” (9 2respondent), “acceptable” (32 respondent). The average of this question is 3.3¼ (82.5%).

Users found that the filters easy to use and understand, by changing the filter for the same word to see how the results would change, they noticed the difference in results which lead to general satisfaction with good average

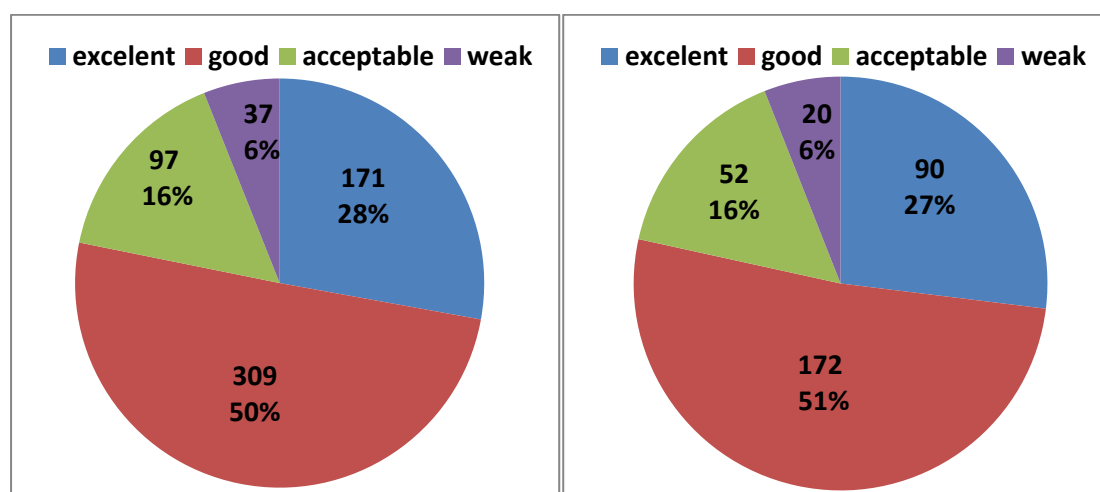


Figure 20. all users opinions about the The helpfulness of the filters.

Figure 21. Users who tried more than one word opinions about the helpfulness of the filters

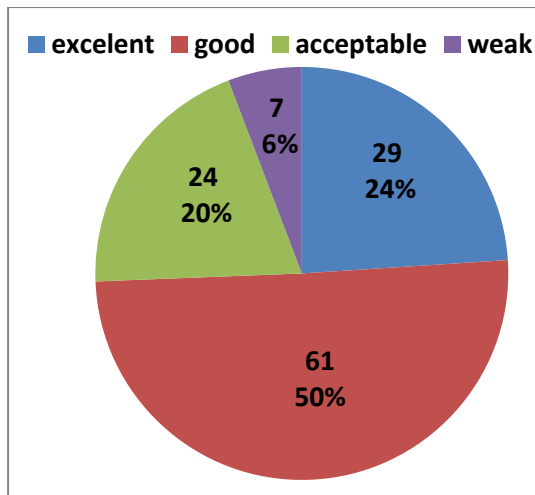


Figure 22. users excluding students opinions About the helpfulness of the filters.

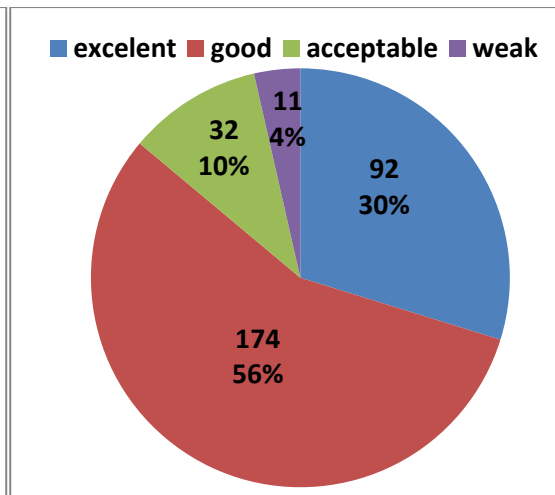


Figure 23. Users looking for translations opinions about the helpfulness of the filters

3.5.12 The efficiency of the results presentation

The respondents were asked whether the presentation and arrangement of the results help to reach the desired results quickly and efficiently, figure 24 shows the answers of all the respondents, most of the respondents answered “good” (291 respondent), while others responded “excellent” (174 respondent), then (127 respondent) and (21 respondent) for “acceptable” and “weak” respectively. The average of this question is $3\frac{3}{4}$ (75%).

Figure 25 shows all the responses excluding the users who answered the option “one word”, the majority answered “good” with (152 respondent), then “excellent” (92 respondent), “acceptable” (71 respondent) and “weak” with (16 respondent). The average of this question is $3\frac{3}{4}$ (75%).

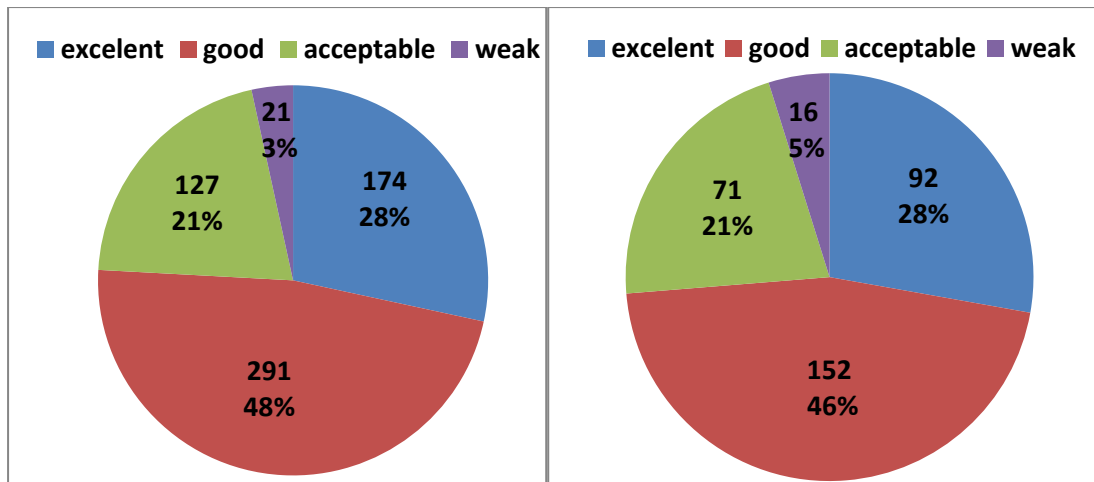


Figure 24. all users opinions about the Efficiency of results presentation.

Figure 25. Users who tried more than one word opinions about the efficiency of results presentation

3.5.13 Linguistic search engine compared to other similar sites

The question focused on comparing the search engine with other similar sites like google translate, the most used one among students, figure 26 shows the answers of all the respondents, where most of the respondents answered “good” (271 respondent), then “excellent” (169 respondent) while few answered “acceptable” (121 respondent) and “weak” (55 respondent). The average of this question is $2.9\frac{1}{4}$ (72.5%)

Figure 27 shows all the responds excluding the users who answered the option “one word”, the majority answered “good” with (147 respondent), then “excellent” (101 respondent), “acceptable” (59 respondent) and “weak” with (26 respondent). The average of this question is $3\frac{1}{4}$ (75%).

Figure 28 shows the responses of all the users excluding the students, to see their opinions regarding the usefulness of the linguistic search engine compared to similar sites, (9 respondent) answered “excellent” and “acceptable” and “excellent” (8 respondent). The average of this question is $3.1\frac{1}{4}$ (77.5%).

Many users relay on google translate in specific to do their assignments or when looking for word translation and synonyms, but after using the Linguistic search engine they found it so rich with resources and how filters help in specifying the results based on their searches, as the average shows it was satisfiable for most users.

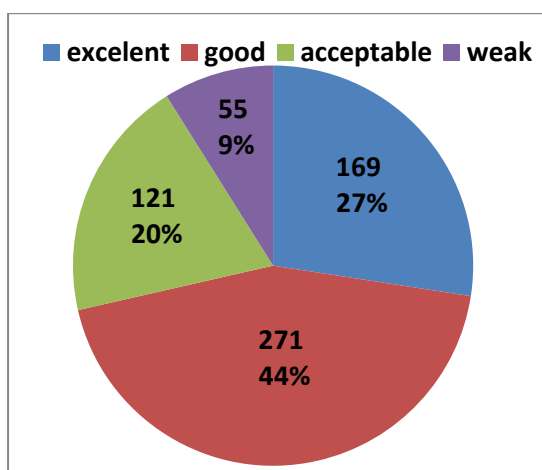


Figure 26. all users opinions about the Comparison between the search engine and similar sites.

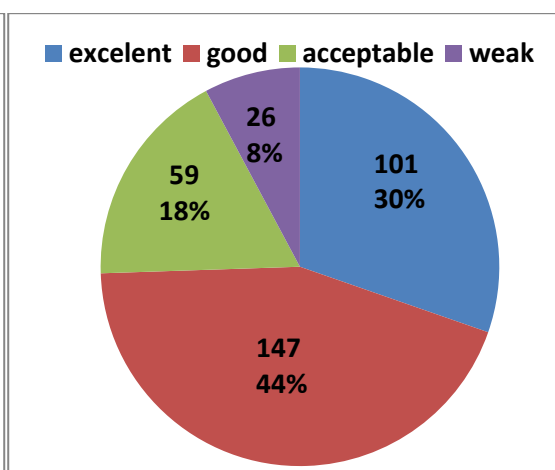


Figure 27. Users who tried more than one word opinions about the Comparison between the Search engine and similar sites

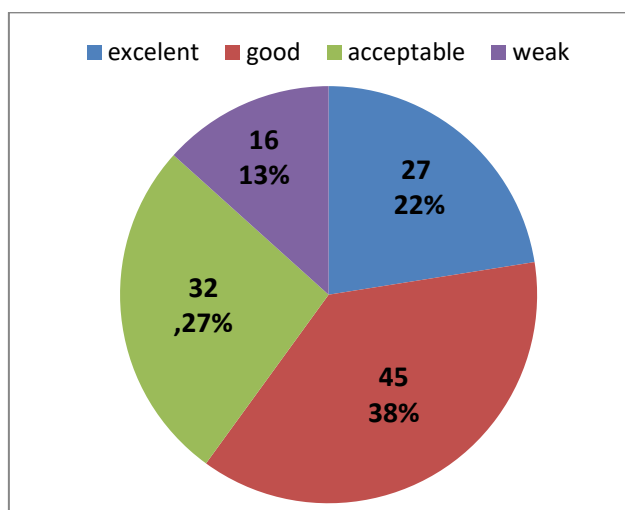


Figure 28. users excluding students opinions about the Comparison between search engine and similar sites.

3.5.14 The usefulness of the results

The respondents were asked if the acquired results are useful and sufficient as desired, figure 29 shows the answers of all the respondents, where many answered “good” (314 respondent) and “excellent” (191 respondent), and few answered “acceptable” (83 respondent) and “weak” (27 respondent). The average of this question is $3\frac{1}{4}$ (75%).

Figure 30 shows all the responds excluding the users who answered the option “one word”, the majority answered “good” with (166 respondent), then “excellent” (105 respondent), “acceptable” (46 respondent) and “weak” with (15 respondent). The average of this question is $3\frac{1}{4}$ (75%).

Figure 31 shows only the responses of the students, to see if they found the results sufficient, the majority answered “good” with (283 respondent), then “excellent” (183 respondent) and “acceptable” (80 respondent). The average of this question is $3.2\frac{1}{4}$ (80%).

Figure 32 shows the responses of all the users excluding the students, to see their opinions whether the results are sufficient, the majority answered “good” with (65 respondent), then “acceptable” (26 respondent) and “excellent” (19 respondent). The average of this question is $2.88\frac{1}{4}$ (72%).

Figure 33 shows only the response of the users who chose the option “looking for translation” as their aim of using the linguistic search engine, the majority answered “good” with (174respondent), then “excellent” (9 2respondent), “acceptable” (32 respondent). The average of this question is $3.2\frac{1}{4}$ (80%).

Figure 34 shows only the response of the users who chose the option “looking for meanings and definitions” as their aim of using the linguistic search engine, the

majority answered “good” with (231 respondent), then “excellent” (148 respondent), “acceptable” (65 respondent). The average of this question is 3.1\4 (77.5%).

Figure 35 shows only the response of the users who chose the option “looking for synonyms” as their aim of using the linguistic search engine, the majority answered “good” with (76 respondent), then “excellent” (53 respondent), “acceptable” (37 respondent). The average of this question is 3\4 (75%).

For this criterion in specific and to see if the results are useful and sufficient for different types of users and also for different search goals, the data was analyzed separately and it seems that for student, as their average is high, the results were sufficient, but for the other types of users their average is lower than the students, the results for them could be not enough based on their searching field or aim, but the overall average is satisfiable.

Furthermore, as for the search goals the averages came very close, which means that the general results or satisfiable for different types of users.

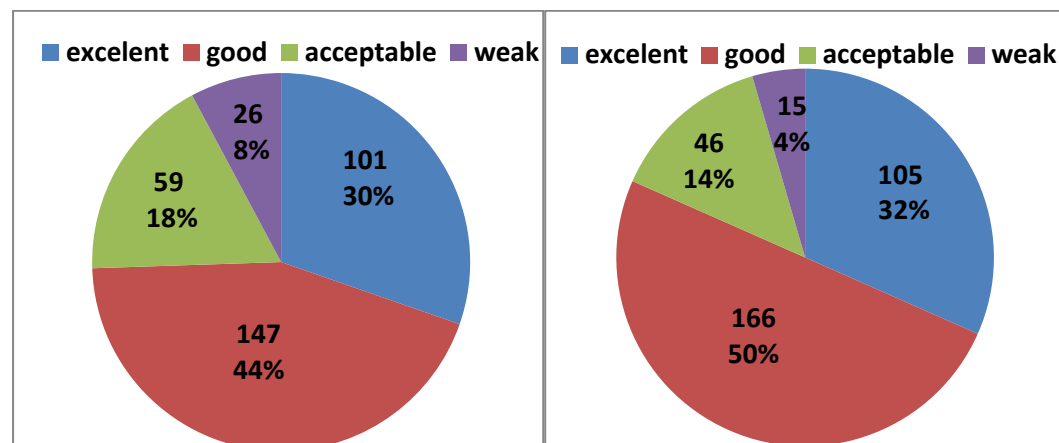


Figure 29. all users opinions about the Usefulness of result.

Figure 30. Users who tried more than one word opinions about the usefulness of results.

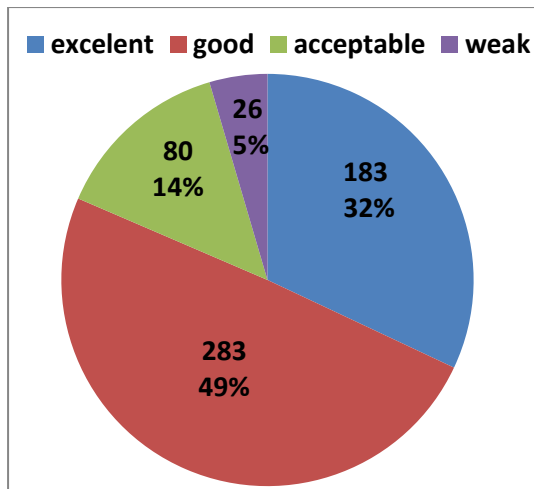


Figure 31. students opinions about the Usefulness of result.

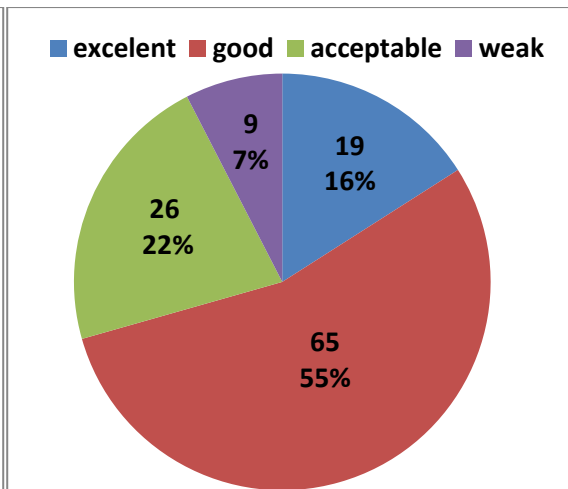


Figure 32. Users excluding students opinions about the usefulness of results.

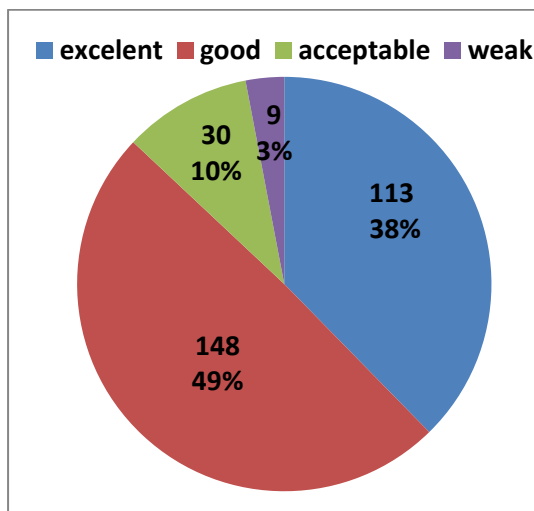


Figure 33. users looking for translations opinion about the usefulness of results

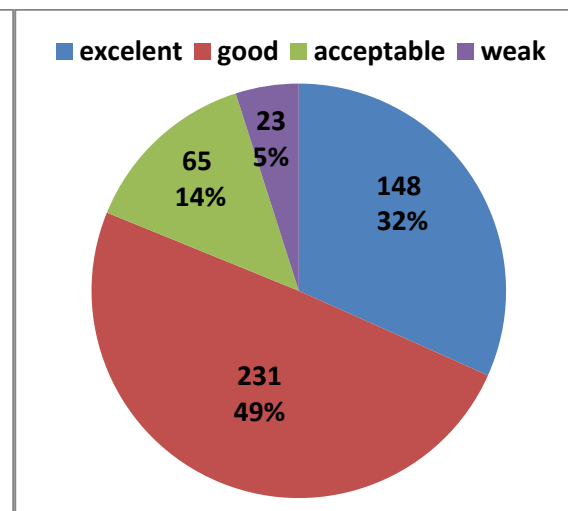


Figure 34. Users looking for definitions opinions about the usefulness of results

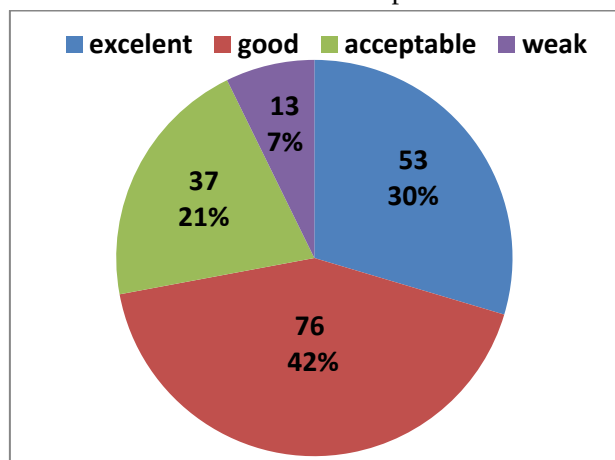


Figure 35. users looking for synonyms opinions about Usefulness of result

3.5.15 Introducing the lexicons

The respondents were asked whether presenting the lexicon name after each result is useful, figure 36 shows the answers of all the respondents, many answered “good” (261 respondent) and “excellent” (242 respondent), and few answered “acceptable” (87 respondent) and “weak” (22 respondent). The average of this question is 3.2\4 (80%)

Figure 37 shows all the responds excluding the users who answered the option “one word”, the majority answered “excellent” with (141 respondent), then “good” (130 respondent), “acceptable” (46 respondent) and “weak” with (14 respondent). The average of this question is 3.2\4 (80%).

Users found it very useful to know the resource of each result, in that way they would have knowledge about the different lexicons that some were digitized especially for the linguistic search engine, and that is shown by the high average of the question.

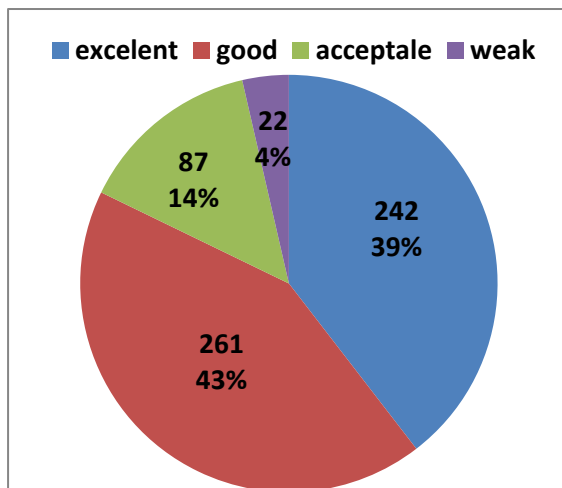


Figure 36. all users opinions about the Resource of the results.

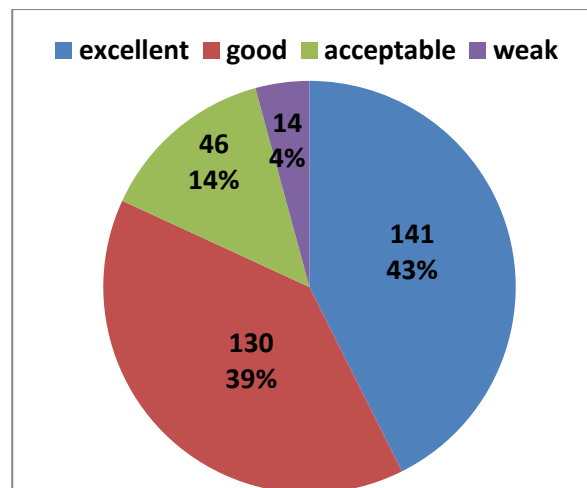


Figure 37. Users who tried more than one word opinions about the resource of results.

3.5.16 Respondents satisfaction

The respondents were asked to rate their experience and whether they would use the Linguistic search engine again, figure 38 shows the answers of all the respondents, most gave high rates, like 4 (34%) , 3 (27%) and 5 (23%). While few low rates were given like 1 (9%) and 2 (7%). The average of this question is 3.6/4 (90%).

The high average of the user satisfaction gives a really good indication that in general the users were satisfied with the results and different features available, thus they would use it again.

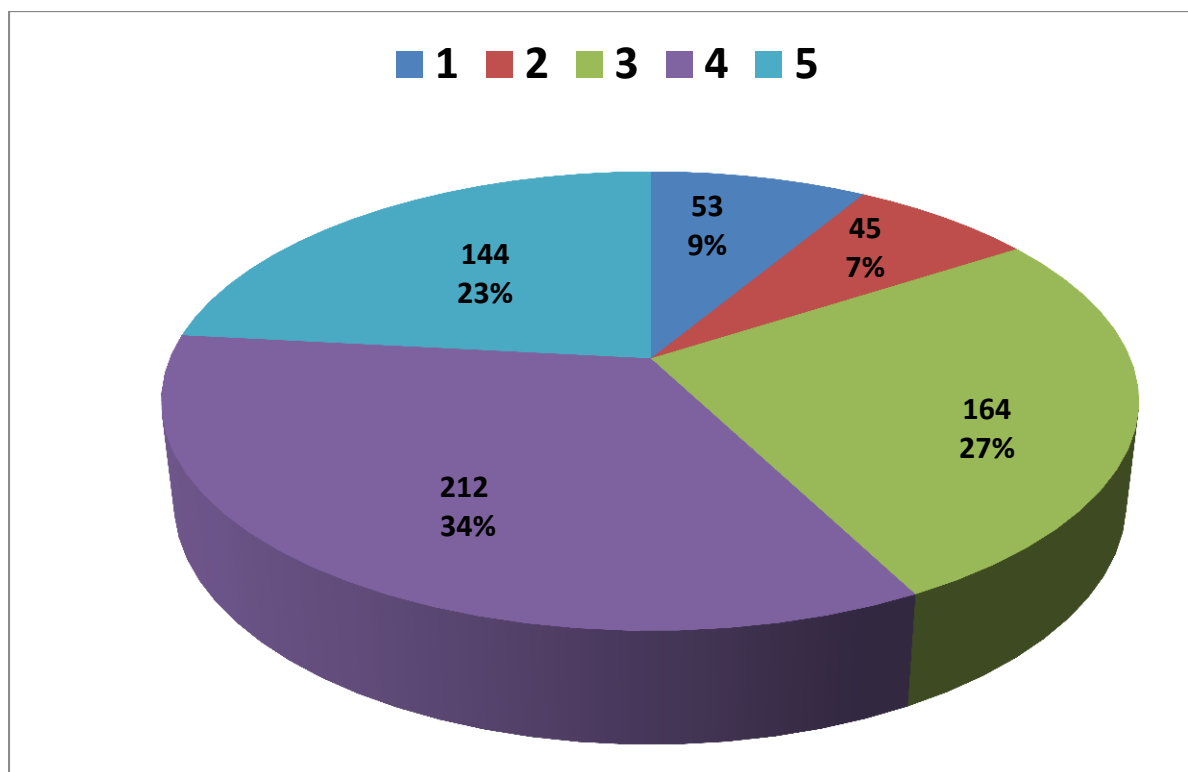


Figure 38. satisfaction rate of all the users

3.6 Summary of the questions and averages

No	Question	Avg	Avg (more than 1 search)	Criteria	
1	Are the fonts and colors of the search engine attractive and comfortable?	73%	73.5%	Design	
2	Are the presentation and the arrangement of the results comfortable?	73%	70.75%	Design	
3	Is the amount of results in a single page comfortable?	75%	72%	Design Effectiveness	
4	Is the linguistic search engine easy to use and learn?	83%	83%	Learnability Efficiency	For students only 83%
5	Do you find it easy to understand the functionality of the filters?	75%	75%	Learnability Effectiveness	For students only 83% For users only looking for translations 80%
6	Is the speed of the linguistic search engine acceptable?	82.5%	85%	Efficiency	
7	Do you find the filters helpful in reaching the desired results quickly and efficiently?	75%	75%	Efficiency	Users excluding students 72.5% For users only looking for translations 82.5%
8	Are the presentation and the arrangement of the results helpful in reaching the desired results quickly and efficiently?	75%	75%	Efficiency	

9	Do you find the linguistic search engine more accurate and useful compared with similar sites like google translate?	72.5%	75%	Effectiveness	Users excluding students 77.5%
10	Do you find the amount of the results sufficient and useful?	75%	75%	Effectiveness	For students only 80% Users excluding students 72% For users only looking for translations 80% For users only looking for definitions 77.5% For users only looking for synonyms 75%
11	Do you find it useful to present the lexicon name with each result?	80%	80%	Effectiveness Learnability	
12	Would you use the linguistic search engine again?	90%		Effectiveness	

Table 2. Summary of the results and averages

3.7 *Controlled Experiment*

Usability criteria, effectiveness and efficiency, of the Linguistic Search Engine were evaluated in a controlled environment, with comparison to Google Translate. Two controlled experiments were carried out in two different universities, Birzeit and Khadoori, where each experiment consisted of 6 participants, thus the total Participants were 12. The aim of this experiment was to measure efficiency, through the metric "Task time" and effectiveness, through the metrics "Task Completion" and "Error frequency" according to the study (García, F.J. et al, 2007).

To start the experiment, two groups were asked to participate in the experiment and formed groups (A and B), each with three students. It was conducted in a lab, using Windows PCs. Group A was assigned to interact with Google Translate and group B was assigned to interact with the Linguistic Search Engine. Group B was given extra time before the experiment in order to learn the basic functionalities and gain knowledge about the website by searching for 10 words and see the results, as it was their first time using the website.

A list of eight tasks was formed considering variant types of questions including looking for translations, definitions, synonyms and semantic difference between terms. Then the task list was distributed to the students in a hardcopy form and each task along with its purpose was explained. Group A was given the tasks to accomplish using Google Translate, while Group B was in parallel doing the same using the Linguistic Search Engine. Furthermore they were asked to deliver their solution of each task in a separate paper, and cannot go to the next question before delivering this paper. Also the time of each task was tracked during the experiment and their interaction within the group was observed to record any errors made. After

finishing the experiment, a comparison between the results of each group was evaluated.

3.7.1 Task time Calculation

The time of each task for each student was recorded in table 3 to measure the efficiency factor of usability, where the first six students were assigned with the Linguistic search engine; while the other 6 students were assigned with Google translate. Additionally according to (Yuan, X. et al, 2015) the mean value for each task is calculated and presented in table 4. Furthermore, in the figure 40 below the medium time of the tasks is represented based to (García, F.J. et al, 2007) and its obvious how Linguistic search engine is more efficient than google translate as the medium time is lower for most questions, also t-test was performed where the mean for all the students of each task on each website was calculated and p value was figured out. Where ($p = 0.276051$) which is statistically significant.

The tasks with long period of time compared with other tasks suggest that they were particularly difficult for the participants or they spent more time thinking and looking for the answer. However, regarding task 6, task 7 and task 8, it can be noticed the participants who used Linguistic search engine took more time comparing to the participants who used Google translate because, the later website did not provide any results, thus the time interval of the search was short.

User		Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8
Linguistic Search engine	User 1	0.58	0.27	0.40	0.20	0.45	1.20	1.56	1.35
User 2		0.43	0.17	1.02	0.30	0.53	0.50	1.56	1.58
User 3		0.30	0.24	0.37	0.51	0.50	1.53	1.50	1.36
User 4		1.50	0.39	1.00	0.31	0.25	3.15	2.03	1.25
User 5		1.56	0.48	1.17	0.43	0.27	3.10	2.00	1.17
User 6		0.50	0.40	1.07	0.43	0.37	2.40	2.30	1.42
Google translate	User 1	1.30	0.25	0.50	0.30	1.01	1.25	1.00	1.10
User 2		2.00	0.14	2.00	1.01	0.50	2.08	1.00	0.30
User 3		1.50	0.30	1.20	1.00	1.11	3.40	1.00	0.30
User 4		0.27	0.26	0.40	0.15	0.18	1.15	0.11	0.12
User 5		0.37	0.27	0.43	0.11	0.22	1.25	0.17	0.30
User 6		0.14	0.20	0.44	0.30	0.30	1.30	0.30	0.11

Table 3. Task efficiency - time (in seconds) by user by task

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8
Linguistic search engine	0.98	0.332	0.837	0.362	0.39	1.958	1.868	1.353
Google translate	1.282	0.32	0.923	0.618	0.619	1.832	0.628	0.428

Table 4. Mean time (in seconds) for each task

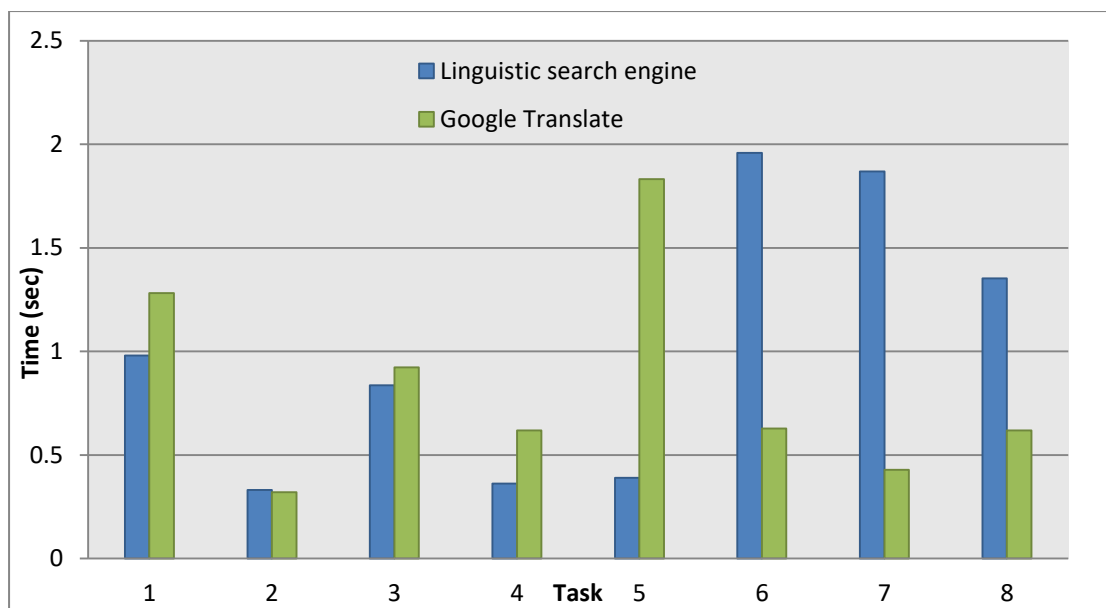


Figure 39. medium time of the tasks on both websites

3.7.2 Task completion comparison

The accuracy of results is measured by relatively comparing the results of each task to a predefined answer sheet by how close the answers compared to the correct one, the results shown in table 5. Also the average of each task was calculated for both websites s shown in table 5. Each task required to provide two different answers if possible, thus in order to measure the accuracy of each task, a metric level was developed, is a way that:

Both answers correct, then grade = 10

One answer correct and the other correct in different context, then grade = 8

Both answers correct in different context, then grade = 6

One answer correct and the other wrong, then grade = 4

One answer correct in different context and the other wrong, then grade = 2

Both answers wrong, then grade = 0

Regarding google translate results, when the point =0 that means that google translate did not provide any valid results or that the answer provided by the participant is not correct.

As shown in figure 30, the averages of results for both websites are compared and obviously the Linguistic search engine results are more accurate as the average is higher for most of the tasks, and the total average is much higher $7.249875 > 3.83325$

User		Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8
Linguistic Search engine	User 1	8	10	10	10	10	8	10	10
User 2		6	10	4	8	10	6	6	8
User 3		10	10	10	10	0	8	0	6
User 4		10	10	0	10	8	6	4	8
User 5		8	10	0	4	0	10	6	10
User 6		8	10	8	8	0	8	6	8
Google translate	User 1	0	0	6	4	10	8	0	0
User 2		8	0	6	0	10	10	0	0
User 3		8	0	10	0	8	8	0	0
User 4		8	0	2	0	10	8	0	0
User 5		4	0	4	4	10	6	0	0
User 6		10	0	4	0	10	8	0	0

Table 5. Task effectiveness – the accuracy of results by user by task

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Total average
Linguistic search engine	8.333	10	5.333	8.333	4.667	7.667	5.333	8.333	7.249875
Google translate	6.333	6.667	5.333	4	9.667	8	0	0	3.83325

Table 6. Average value for each task

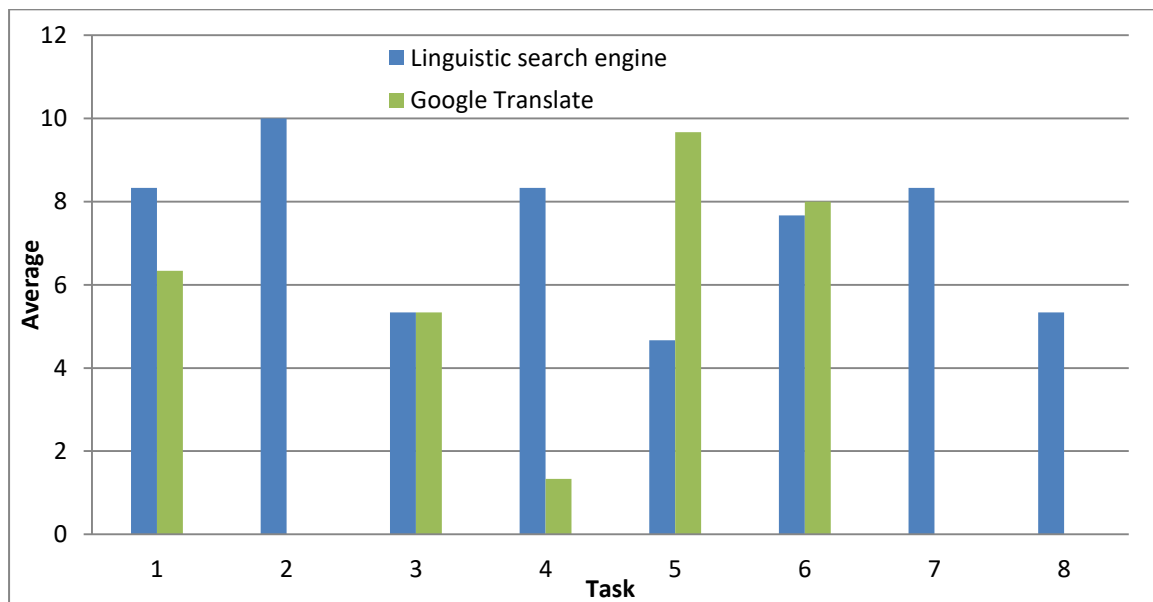


Figure 40. Tasks Average on both websites

3.7.3 Error frequency comparison

Most participants applied for both websites faced the issue of comprehending the required tasks, like when the task requires synonyms, the participants could be looking for definitions. Other type of errors is when the task requires translation to Arabic and the participants provide definition in the same language especially faced by the Google translating participants. These were the main source of confusion and errors faced during the experiment.

Results and Discussion

3.8 Results

This subsection presents the results obtained from the analysis of the results obtained from the questionnaire regarding the usability evaluation of Birzeits Linguistic search engine. the targeted population was from different universities and backgrounds to get a wider spectrum of opinions and feedback. They have expressed what they saw a downside in the general design, such as adding more colors in the background, offer more languages, to make a mobile application version of it, offer text to speech functionality, remove the ontology box from the main page, add pictures to represent the results and many more feedback...

On the other hand, from the functionality side, they have found it extraordinary that so much work and effort have been put to digitize 150 dictionaries over the course of 8 years and also when it comes to incorporate filters for achieving better results.

Furthermore table 1 show that generally the users were satisfied with the usability of the linguistic search engine based on the Likert scores in the questionnaire which could be summarized as follows:

- Design: the first three questions in the table were set to measure the design criterion, in terms of how comfortable the user feels about the fonts, colors, the presentation and arrangement of the results and the amount of results in a single page. The average of those questions was about 73%
- Learnability: the following two questions were set to measure the learnability criterion, based on the users first interaction with the search engine and

whether they found it easy to use and learn, and also understand the functionality of the existing filters. The average of those questions was 83% especially for the students as their average was high.

- Efficiency: the following three questions were set to measure the efficiency criterion, based on the speed of the linguistic search engine and how the filters and the presentation of the results help the users reach the desired results quickly. The average of those questions was about 75% and especially for the students as their average was high.
- Effectiveness: the last four questions were set to measure the effectiveness criterion, based on the its accuracy compared with similar sites like Google Translate, whether the amount of results was sufficient, also whether introducing the lexicon sources of the results was helpful to the users and lastly how satisfied they were the search sites and if they would use it again. The average of those questions was higher than the other criteria, about 80%.

Table 6 shows a summary of the averages of each criterion, even though it does not reflect the accuracy of the usability of the Linguistic search engine, but it gives a total close overview.

Criterion	Design	Learnability	Efficiency	Effectiveness	Average
Average	73%	83%	75%	80%	77.8%

Table 7 summary of averages for each criterion

3.9 Discussion

The results of the study show that different usability factors were studied. The general usability of the Linguistic search engine is very satisfiable, as there are many types of users each with different background, experience and personal perspectives.

Also the usability of many questions would not be accurate when considering the answers of the people who tried one word or less than five words as their results would not be reliable. Thus the average was calculated twice.

To discuss more in detail the questions with the lowest averages:

- Fonts and colours of the search engine: although many users expressed that using blue colour for the results indicate that it is clickable while it is not, but the general design is subjective to the user's personal aesthetics, so satisfying more that 73% of the total user can be considered very well.
- The presentation and arrangement of the results: many users expressed that the "ontology box" on the right side of the page is confusing and there is no sense of putting it in the main page. This comes from lack of knowledge among the users in the meaning of ontology and how it would help them achieve better and accurate results. Nonetheless the average is good.
- The amount of results in a single page: different opinions regarding this criterion either the results are too much for a single page or it is enough. Comparing with all the search engines and website it is considered that showing more than ten results in a single page is not usable and desirable. Thus the search engine shows ten results which is just the right amount.

- The accuracy of the search engine compared with similar sites: to measure the accuracy right, it would be wrong to take into account the users who tried one word or less than five words as their results are neither subjective nor reasonable. Thus considering the average of the other users gives a better perspective.
- Whether the results are sufficient and useful: the average of student was high indicating that they found the results sufficient as commonly they get enough results from the first two to three pages, while researchers and writers would go through all the pages when looking carefully for the desired results. Thus the difference in the averages for both types of users. Also for the different search goals it is concluded that when it comes for looking for translations. Users were satisfied most with the amount of results.

3.10 Recommendations for usability improvements

- Fonts and colours can be improved based on the usability heuristics of search engines.
- The “ontology box” can be removed from the main page, as to not confuse the users, but it would help to introduce the meaning of ontology to them to gain better results.
- Constricting ten results in a single page to conform to the general design heuristics.

3.11 *Limitations*

- Before conducting this study, the evaluator had limited experience with usability testing. If it was the case with an experienced evaluator, it is possibly that a more effective test have been designed and more usability problems would be revealed.
- The sample size of 622 participants can be considered more than sufficient, but the majority of respondents were students, therefore there should more different types of respondents to increases the reliability of the results.
- There was not enough time to include different types of respondents in a course of one semester, as this needs more than one evaluator and more time to include a bigger sample.

3.12 *Future work*

- To get better results, users who interacted with the search engine more and tried more than 20 words can better analyze and give better feedback and ways to improve better.
- The questionnaire should target a wider spectrum of different users such as: hard working mothers who would benefit from the search engine when teaching their children, teachers, lawyers, journalists, engineers, linguists, University lecturers, school students
- Text- to- Speech feature could be included in the search engine. To help the user with the right pronunciation of words.

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Appendix 1

Controlled experiment tasks

Task 1: Giving the following sentence, suggest two other words (i.e., synonyms) that can be used to replace the word **Elaborate**, without changing the meaning of the sentence:

The main idea of this lesson is elaborated in the textbook

Solutions:

Word 1:

Word 2:

Task 2: Giving the following sentence, suggest two other words (i.e., synonyms) that can be used to replace the word (صيرورة), without changing the meaning of the sentence:

صيرورة العملية الديمقراطية تعمل بقوة وزخم في اللحظة الراهنة

Solutions:

Word 1:

Word 2:

Task 3: Giving the following sentence, what would be the meaning of the the word **account**:

This report does not provide a sufficient account for constructing a hospital in this area.

Solutions:

meaning 1:

meaning 2:

Task 4: What is the meaning of (رتاج) in the following statement

لم يعجبه رتاج القصر

Solutions:

meaning 1:

meaning 2:

Task 5: Provide two English translations for the following Arabic Statement (please translate it yourself, and don't use machine translation or provide a literal translation):

التقيت بهذا الرجل المتعطر

Solutions:

Translation 1:

Transition 2:

Task 6: Provide two Arabic translations for the following English Statement (please translate it yourself, and don't use machine translation or provide a literal translation):

The proliferation of technology has limited our interaction and exposure to nature

Solutions:

Translation 1:

Transition 2:

Task 7: what is the semantic difference between (الرزانة) and (الوقار)

Solutions:

Meaning of (الرزانة):

Meaning of (الوقار):

The difference:

Task 8: what is the semantic difference between (أعجمي) and (عجمي)

Solutions:

Meaning of (أعجمي):

Meaning of (عجمي):

The difference: